S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Aluminium Hydroxide** **Label Claim:** **475mg/5ml**	**METHOD 1:** **Procedure:** Mix a quantity of the oral suspension equivalent to 5mL of syrup with 30 mL of 2M *hydrochloric acid* and add sufficient *water* to produce 100 mL.Dilute 10 mL of the solution obtained with 20 mL of *water*. Add 25 mL of 0.05M *disodium edetate VS*, mix, allow to stand for 5 minutes and add 20 mL of *acetate buffer pH 4.4*, 60 mL of *ethanol (96%)* and 2 mL of a 0.026% w/v solution of *dithizone* in *ethanol (96%)*. Titrate with 0.05M *zinc sulfate VS* until a bright pink colour is obtained. Repeat the procedure without the substance being examined. The difference between the titrations represents the amount of disodium edetate required. Each mL of 0.05M *disodium edetate VS* is equivalent to 3.90 g of $Al(OH)_3$. $\% \, \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ 3.900\ x\ 100}{47.5 \times 0.744}$ **METHOD 2:** **Procedure:**

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

Mix a quantity of the oral suspension equivalent to 5mL of syrup with 30 mL of 2M *hydrochloric acid* and add sufficient *water* to produce 100 Ml. Dilute 20ml of the solution obtained with 40ml of already standardized 0.05M EDTA and add 80ml of DM. Add 2 to 3 drops of Methyl red (neutralized with 0.1M NaOH). Heat in water bath for 30 mins and allow to cool. Add 3g of Hexamine and about 50mg of Xylene – Orange titurant. Titrate against 0.05M Lead Nitrate.

Each ml of 0.05M EDTA is equivalent to 2.549mg of Al_2O_3.

$$\% \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ 2.549\ x\ 100}{95 \times 0.744}$$

METHOD 3:

Reagents:

Ammonium Acetate-Acetic Acid Buffer Solution

Weigh about 7.71g of ammonium acetate into 100ml volumetric flask.

Add 40ml of de-ionized water, cover and shake till dissolution is complete.

Add 5.7ml of acetic acid, dilute to the mark with de-ionized water and mix

Dithizone Solution

Weigh about 25.6mg of dithizone into 100ml volumetric flask.

Add 40ml of ethanol, cover and shake till dissolution is complete.

Dilute to the mark with de-ionized water and mix

0.05M Zinc Sulphate Solution

Accurate weigh and transfer about 14.4g of zinc sulphate powder into a 1-litre volumetric flask.

Add 500ml of de-ionized water; mechanically shake/sonicate till dissolution is complete

Dilute to the mark with de-ionized water and mix

Standardize as follows:

Pipette 20.0ml of each of 0.05M Disodium Edetate into two separate 150ml conical flask or beaker.

Add in the following order with continuous mixing:

Add o each, 50ml of ethanol and 2ml Dithizone Solution.

Titrate with 0.05M Zinc Sulphate to a bright rose-pink color end-point.

$$\text{Molarity of } ZnSO_4 = \frac{20 \times 0.05}{\text{Titre}}$$

Sample Solution Preparation

Transfer 5ml of suspension accurately measured and previously well shaking

Add 10ml of water stir and slowly add 5ml of conc. Hcl, warm gently to dissolve

Dilute to about 50 ml, cool and filter into 100ml volumetric flask wash the filter paper with water and collect the filtrate into the same flask

Add water to volume and mix

Procedure

Pipette 10ml of sample solution above into 250ml conical flask

Add 20ml of water

Then add with continuous stirring 25ml of disodium edetate (0.05M)

Add 20ml of acetic acid ammonium acetate buffer Ts

And heat to near the boiling temperature for 5minutes

Cool and add 50ml of absolute ethanol

Add about sample 4ml, blank 6ml of dithizole (0.0265) and mix

Titrate with 0.05M $ZnSO_4$ Vs until colour changes to pink

Perform a blank determination by substituting 10ml of water for d assay preparation

Calculation

% Al_2O_3

$$= \frac{\text{(Blank Titre – Sample Titre) x Actual Molarity of } ZnSO_4 \text{ x 0.002549g x 100 x 100}}{\text{0.475 (g) x 0.05 x 5}}$$

3.	**Magnesium Hydroxide** **Label Claim:** **100mg/5ml**	**METHOD 1:** Mix a quantity of the oral suspension equivalent to 5mL of syrup with 30 mL of 2M *hydrochloric acid* and add sufficient *water* to produce 100 mL. Dilute 10 mL of this solution to 200 mL with *water*. Add, while stirring, 1 g of *ammonium chloride*, 20 mL of *triethanolamine*, 10 mL of *ammonia buffer pH 10.9* and 0.4 mL of *mordant black 11 solution*. Titrate with 0.05M *disodium edetate VS* until a blue colour is obtained. Repeat the procedure without the substance being examined. The difference between the titrations represents the amount of disodium edetate required. Each mL of 0.05M *disodium edetate VS* is equivalent to 2.916 g of Mg $(OH)_2$. $\% \, \boldsymbol{Assay} = \frac{(\text{Average Titre} \times 2) \times \text{Standard Factor} \times 2.916 \times 100}{20}$ ***REFERENCE:*** *British Pharmacopoeia, 2024, Ammonium Chloride: Ph. Eur. Monograph 0007.*
4.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the

sample.

- Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an “S” or “8” shape.
- Allow the media to completely solidify.
- Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\text{CFU/mL} = \frac{\text{Total number of colonies obtained x dilution factor}}{\text{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats.
- The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105**
- If the colonies are fused or the whole plate is covered with a single colony, then report

		as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REV	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		